

TUTORIAL - 7 :-

① Find the directional derivative of $x^2+y^2+z^2=9$ in the direction of the vector $x^2z\hat{i}+xy^2\hat{j}+yz^2\hat{k}$ at the point $(2,2,1)$.

Soln:- Let $\phi = x^2+y^2+z^2-9$ be the given surface.

We know that,

$$\nabla\phi = \frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k}$$

$$\nabla\phi = 2x\hat{i} + 2y\hat{j} + 2z\hat{k}$$

At point $(2,2,1)$,

$$\nabla\phi = 4\hat{i} + 4\hat{j} + 2\hat{k} \quad \text{--- (1)}$$

Also, let $\vec{e} = x^2z\hat{i} + xy^2\hat{j} + yz^2\hat{k}$ be the given vector

$$\therefore \hat{e} = \frac{\vec{e}}{|\vec{e}|} \text{ at point } (2,2,1)$$

$$\hat{e} = \frac{4\hat{i} + 8\hat{j} + 2\hat{k}}{\sqrt{16+64+4}}$$

$$\hat{e} = \frac{1}{2\sqrt{21}}(4\hat{i} + 8\hat{j} + 2\hat{k}) \quad \text{--- (2)}$$

Directional

Derivative $D\phi = \nabla\phi \cdot \hat{e}$

$$= (4\hat{i} + 4\hat{j} + 2\hat{k}) \cdot \frac{1}{2\sqrt{21}}(4\hat{i} + 8\hat{j} + 2\hat{k}) \quad (\text{from (1,2)})$$

$$= \frac{16+32+4}{2\sqrt{21}}$$

$$\boxed{D\phi = \frac{26}{\sqrt{21}}}$$

② show that vector field $\vec{F} = (x^2 - yz)\hat{i} + (y^2 - xz)\hat{j} + (z^2 - xy)\hat{k}$ is irrotational.

Soln: Given $\vec{F} = (x^2 - yz)\hat{i} + (y^2 - xz)\hat{j} + (z^2 - xy)\hat{k}$ — (1)

We know that,

$$\text{curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2 - yz & y^2 - xz & z^2 - xy \end{vmatrix}$$

$$= \hat{i}(-x - (-x)) - \hat{j}(-y - (-y)) + \hat{k}(-z - (-z))$$

$$= \hat{i}(0) + \hat{j}(0) + \hat{k}(0)$$

$$\boxed{\text{curl } \vec{F} = \vec{0}}$$

Hence $\vec{F}$ is irrotational.

③ Find $\text{curl}(\text{curl } \vec{A})$, where $\vec{A} = x^2y^2\hat{i} - 2xyz\hat{j} + 2y^2z\hat{k}$ at the point $(1, 1, 2)$.

Soln: Given $\vec{A} = x^2y^2\hat{i} - 2xyz\hat{j} + 2y^2z\hat{k}$

We know that,

$$\text{curl } \vec{A} = \nabla \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2y^2 & -2xyz & 2y^2z \end{vmatrix}$$

$$= \hat{i}(4yz + 2xy) - \hat{j}(0 - 0) + \hat{k}(-2yz - 2x^2y)$$

$$\text{curl } \vec{A} = (4yz + 2xy)\hat{i} - (2yz + 2x^2y)\hat{k}$$

$$\text{Now, } \text{curl}(\text{curl } \vec{A}) = \nabla \times (\nabla \times \vec{A}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 4yz + 2xy & 0 & -(2yz + 2x^2y) \end{vmatrix}$$

$$= \hat{i}(-2z + 2x^2) + \hat{j}(4xy + 4y) + \hat{k}(-4z - 2x)$$

At point $(1, 1, 2)$,

$$\boxed{\text{curl}(\text{curl } \vec{A}) = -6\hat{i} - 8\hat{j} - 10\hat{k}}$$

4) show that $\vec{F} = (\sin z + y)\hat{i} + x\hat{j} + x\cos z\hat{k}$ is irrotational.
Also find a scalar function ϕ such that $\vec{F} = \nabla\phi$.

Soln: Given $\vec{F} = (\sin z + y)\hat{i} + x\hat{j} + x\cos z\hat{k}$ — (1)

We know that,

$$\text{curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \sin z + y & x & x\cos z \end{vmatrix}$$

$$= \hat{i}(0 - 0) - \hat{j}(\cos z - \cos z) + \hat{k}(1 - 1)$$

$$\text{curl } \vec{F} = \hat{i}(0) + \hat{j}(0) + \hat{k}(0)$$

$$\boxed{\text{curl } \vec{F} = \vec{0}}$$

$\therefore \vec{F}$ is an irrotational vector.

Also, $\vec{F} = \nabla\phi$

$$(\sin z + y)\hat{i} + x\hat{j} + x\cos z\hat{k} = \hat{i}\frac{\partial\phi}{\partial x} + \hat{j}\frac{\partial\phi}{\partial y} + \hat{k}\frac{\partial\phi}{\partial z}$$

on comparing,

$$\frac{\partial\phi}{\partial x} = \sin z + y, \quad \frac{\partial\phi}{\partial y} = x, \quad \frac{\partial\phi}{\partial z} = x\cos z$$

$$\phi = x\sin z + xy + f(y, z), \quad \phi = xy + f(x, z), \quad \phi = -x\sin z + f(x, y)$$

$$\therefore \boxed{\phi = x\sin z + xy}$$

5) If $\vec{F} = \nabla(xy^3z^2)$, find $\text{div } \vec{F}$ and $\text{curl } \vec{F}$ at point $(1, 1, 1)$.

Soln: Given $\vec{F} = \nabla(xy^3z^2)$

$$\text{We know that, } \nabla\phi = \hat{i}\frac{\partial\phi}{\partial x} + \hat{j}\frac{\partial\phi}{\partial y} + \hat{k}\frac{\partial\phi}{\partial z}$$

$$\vec{F} = \nabla\phi = y^3z^2\hat{i} + 3xy^2z^2\hat{j} + 2xy^3z\hat{k}$$

$$\text{div } \vec{F} = \nabla \cdot \vec{F}$$

$$= \frac{\partial}{\partial x}f_1 + \frac{\partial}{\partial y}f_2 + \frac{\partial}{\partial z}f_3$$

at point $(1, 1, 1)$,

$$\text{div } \vec{F} = 0 + 6xy^2z^2 + 2xy^3z$$

$$\text{div } \vec{F} = -6 - 2 = \boxed{-8}$$

$$\text{curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 z^2 & 3xy^2 z^2 & 2xy^3 z \end{vmatrix}$$

$$= \hat{i} (6xy^2 z - 6xy^2 z) - \hat{j} (2y^3 z - 2y^3 z) + \hat{k} (3y^2 z^2 - 3y^2 z^2) \\ = \hat{i}(0) + \hat{j}(0) + \hat{k}(0)$$

$$\boxed{\text{curl } \vec{F} = \vec{0}} \Rightarrow \text{vector } \vec{F} \text{ is irrotational.}$$

⑥ If $\vec{F} = 2x^2 yz \hat{i} - 3x^2 z \hat{j} + 5y^2 z \hat{k}$, find $\text{div } \vec{F}$, $\text{curl } \vec{F}$, $\text{div}(\text{curl } \vec{F})$ at $(-1, 2, -3)$.

Solu: Given $\vec{F} = 2x^2 yz \hat{i} - 3x^2 z \hat{j} + 5y^2 z \hat{k}$

We know that, $\text{div } \vec{F} = \nabla \cdot \vec{F}$

$$= \frac{\partial}{\partial x} f_1 + \frac{\partial}{\partial y} f_2 + \frac{\partial}{\partial z} f_3$$

$$\text{div } \vec{F} = 4xyz + 0 + 5y^2$$

At point $(-1, 2, -3)$,

$$\text{div } \vec{F} = 24 + 20 = \underline{\underline{44}}$$

Also, $\text{curl } \vec{F} = \nabla \times \vec{F}$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x^2 yz & -3x^2 z & 5y^2 z \end{vmatrix}$$

$$\text{curl } \vec{F} = \hat{i} (10yz + 3x^2) - \hat{j} (0 - 2x^2 y) + \hat{k} (-6xz - 2x^2 z)$$

At point $(-1, 2, -3)$,

$$\text{curl } \vec{F} = \underline{\underline{-57\hat{i} + 4\hat{j} - 12\hat{k}}}$$

$$\text{Now, } \text{div}(\text{curl } \vec{F}) = \nabla \cdot \text{curl } \vec{F} = \frac{\partial}{\partial x} f_1 + \frac{\partial}{\partial y} f_2 + \frac{\partial}{\partial z} f_3$$

$$= 6x + 2x^2 - 6x - 2x^2$$

$$\boxed{\text{div}(\text{curl } \vec{F}) = 0}$$